

From Data Matching to Financial Integrity: An AI-Enhanced Reconciliation Framework for Nepal's Social Security Systems

Architectural Blueprint for a Pragmatic Real-Time Reconciliation Engine

The design of the reconciliation system's core architecture requires a careful balance between the aspirational goal of "real-time" processing and the pragmatic constraints of a development environment reliant on mocked data and a defined timeline . The theoretical ideal of a real-time system would involve an event-driven architecture where new claims from OpenIMIS and corresponding payments from a live SOSYS are processed as discrete events occur [3](#) . This would necessitate robust APIs for both systems and a continuous background service, potentially leveraging a message broker like Apache Kafka to manage the flow of data [30](#) . However, given the explicit acknowledgment in the implementation guide that direct API access to government systems is unlikely and that synthetic data is required, a more practical approach is to simulate real-time capability through a high-frequency batch processing model [13](#) . This strategy provides the observable *effect* of real-time operation—the immediate detection of discrepancies—without the significant architectural overhead of a full-scale streaming platform. The recommended path involves creating a Python-based reconciliation script, encapsulated within a FastAPI service, which can be triggered by a scheduler like cron at very short intervals (e.g., every few seconds or minutes) during demonstrations . This ensures that the reconciliation status is updated almost instantaneously after new data is introduced into the mocked databases, fulfilling the spirit of the real-time requirement.

The foundational components of this architecture are the data sources, the processing engine, and the communication layer. The first source is OpenIMIS, an open-source Insurance Management Information System used to manage health insurance processes, including beneficiary enrollment, claims, and provider management [23](#) [59](#) . For this project, a local instance of OpenIMIS can be spun up using Docker, providing access to its PostgreSQL database via a GraphQL API at `localhost:8000/api/graphql` .

Alternatively, if the Docker setup proves too time-consuming, the schema of the relevant claims table can be replicated in a separate Postgres database or even exported as CSV files, as the reconciliation logic is primarily concerned with the structure of the data rather than its physical location . The second component is the mocked SOSYS dataset, which represents Nepal's Social Security Management Information System, the entity responsible for tracking government payments and disbursements [17](#) . This mocking strategy is not just a fallback but a deliberate choice to maintain control over the data environment and scope of the project . The SOSYS mock should be structured as a relational table mirroring a payment ledger, with fields such as `payment_id`, `claim_id` (to link back to OpenIMIS), `amount_paid`, `payment_date`, and `status` . The generation of this synthetic data is a critical step; it should not be random but should be deliberately seeded with realistic mismatches, such as injecting 10-15% missing payments, partial amounts, duplicates, and wrong amounts, to create a credible demo dataset .

The reconciliation engine itself, as outlined in the preliminary guide, is best implemented in Python for its powerful data manipulation libraries. Using the Pandas library, the engine would perform a left join between the claims DataFrame and the payments DataFrame on the `claim_id` field, using the `_merge` indicator to classify each record . This process yields four primary reconciliation statuses: 1. **RECONCILED**: Records present in both datasets with matching `claim_id` and no further discrepancies. 2. **MISSING_PAYMENT**: Records present in the OpenIMIS claims dataset but absent in the SOSYS payments dataset (`_merge == 'left_only'`). This signifies that a claim has been approved but no corresponding payment has been recorded. 3. **AMOUNT_MISMATCH**: Records found in both datasets, but the `amount_claimed` from OpenIMIS does not equal the `amount_paid` from SOSYS. This highlights underpayments or overpayments. 4. **STATUS_PENDING**: Records found in both datasets, but the status in SOSYS indicates a pending state (e.g., 'pending', 'partial') rather than a final 'paid' status, suggesting a delay in the payment lifecycle.

This rule-based classification forms the deterministic foundation of the system, where matches are determined by explicit criteria, similar to how deterministic patient matching systems use fixed rules based on identifiers like Social Security numbers [3](#) . To expose this logic to the rest of the application, particularly the frontend dashboard, the entire reconciliation function should be wrapped in a FastAPI service. This microservice would contain an endpoint (e.g., `/reconcile`) that, when called, executes the reconciliation script, loads the latest data from the two data sources (OpenIMIS and mocked SOSYS), performs the join and classification, and returns the enriched dataset as a JSON object . This modular approach decouples the data processing logic from the presentation layer, allowing for independent development and future scaling. The suggestion to use

Mojaloop introduces an element of complexity that warrants careful consideration . Mojaloop is a reference architecture for interoperability between financial institutions, designed for modernizing payment systems and enabling low-cost transaction services. While it is possible that SOSYS involves inter-bank fund transfers, integrating Mojaloop at this stage may be premature and could distract from the core objective of building a reliable reconciliation pipeline between OpenIMIS and a mocked payment ledger. The initial focus must remain on validating the reconciliation logic and user-facing features. Only if a specific requirement emerges to simulate complex payment network interactions would the integration of Mojaloop become a justified architectural decision. For now, treating the mocked SOSYS as a simple database table or set of CSVs is the most efficient path to achieving the project's goals.

Component	Technology/Method	Rationale
OpenIMIS Access	Self-hosted Docker instance or replicated schema	Provides controlled access to claim data without requiring public API access .
SOSYS Mocking	Structured PostgreSQL table / CSV files	Mirrors a payment ledger, allowing for realistic data injection and controlled experiments .
Data Generation	Python/Faker with rule-based anomaly injection	Creates a synthetic dataset with realistic patterns and deliberate mismatches (e.g., 10-15%) .
Reconciliation Logic	Python (Pandas) with SQL-like joins	Efficiently classifies records into reconciled, missing, mismatched, and pending statuses .
Processing Model	High-frequency batch jobs (e.g., via cron)	Simulates real-time capability by updating status frequently without a complex streaming architecture.
API Layer	FastAPI	Provides a lightweight, asynchronous interface for the frontend to trigger reconciliation and fetch results .
Mojaloop Integration	Not currently planned	Premature and adds unnecessary complexity; focus remains on the core reconciliation logic.

This architectural blueprint prioritizes functionality and demonstrability within the project's constraints. By simulating real-time behavior through rapid batch processing and focusing on a clean, well-defined API, the system can effectively showcase its core value proposition: the automated and near-instantaneous detection of financial discrepancies between a claims management system and a payment ledger. This foundation is essential before moving to more advanced layers like AI-powered explanations and sophisticated visualizations.

Data Strategy: Fidelity, Synthesis, and Role-Based Explanations

A successful reconciliation system hinges on the quality, realism, and utility of its underlying data. The strategy for handling data must address three interconnected challenges: ensuring the fidelity of the mocked SOSYS dataset, generating meaningful and actionable AI explanations for flagged anomalies, and tailoring those explanations to the specific cognitive models of different end-users. The initial step, as established, is the creation of a synthetic SOSYS dataset . This is not merely about populating tables with random numbers but about imbuing them with domain-specific realism. The data generation process should move beyond simple randomness and adopt a rule-based approach to mimic real-world scenarios. For instance, instead of randomly marking 10% of claims as "missing," one could define business rules such as: "In District Kavrepalanchok, 5% of all claims exceeding NPR 150,000 submitted on the last day of the month are intentionally delayed for budgetary reasons." Such rules can be derived from principles of fraud detection, which often rely on attributional rules to screen claims [2](#) . This method of synthetic data generation enhances the credibility of the entire system; if the AI explanation later flags a claim as anomalous because its amount is five times higher than the hospital's historical average, it reinforces the plausibility of the underlying data used to calculate that average [39](#) .

The next layer of the data strategy is the AI explanation engine. The goal here is to translate the abstract reconciliation status (e.g., 'AMOUNT_MISMATCH') into plain-language insights that are immediately understandable and actionable. The implementation guide correctly identifies a critical pitfall for a live demonstration: making real-time calls to a Large Language Model (LLM) like Claude can introduce unacceptable latency or failure points during a pitch . Therefore, the recommended approach is to pre-generate these explanations for the entire demo dataset. This involves creating a library of parameterized prompts that feed claim-specific details (such as `hospital_name`, `amount_claimed`, `district`) into the LLM. This allows for dynamic personalization while guaranteeing reliability and speed during the presentation. The prompt engineering itself is a nuanced task. The prompt must instruct the LLM not only to explain the anomaly but also to adopt a specific tone and level of detail appropriate for a given user role [39](#) . For example, a prompt for a cashier might emphasize clear, step-by-step instructions, while a prompt for a ministry official would focus on summarizing systemic trends and risks.

This leads to the third and most sophisticated aspect of the data strategy: role-specific customization. Different stakeholders interact with the system for fundamentally different

reasons and require different types of information. The system must cater to these distinct mental models to be truly effective. The preliminary analysis suggests three primary user roles, each with a unique information need:

1. **The Cashier:** At the operational front line, the cashier's priority is to resolve issues quickly and efficiently. Their view must be highly action-oriented. An AI explanation for a 'MISSING_PAYMENT' status for Claim #123 from 'Bir Hospital' might read: "You have a flagged claim for Bir Hospital. The system shows an approved claim for NPR 50,000 dated 2024-10-25, but no payment record exists in the mocked SOSYS. Please verify the manual entry log and confirm if the payment was processed outside the system." This language is direct, uses imperative verbs, and focuses solely on the immediate action required.
2. **The Officer:** As a mid-level analyst, the officer needs more contextual detail to diagnose the root cause of a discrepancy. They are not just resolving a single issue but are part of a larger workflow involving providers and auditors. For the same 'AMOUNT_MISMATCH' anomaly, their explanation could be: "Claim #123 from Bir Hospital shows an 'AMOUNT_MISMATCH'. The approved amount in OpenIMIS was NPR 50,000, but the mocked SOSYS payment record only reflects NPR 40,000. There is a difference of NPR 10,000. You should contact the provider to clarify the basis for the reduced payment and request supporting documentation for the variance." This version includes specific figures, frames the issue as a variance, and suggests a logical next step involving external communication.
3. **The Ministry Official:** Operating at a strategic level, this user is concerned with aggregate performance, policy effectiveness, and systemic risks. Their explanation should synthesize individual anomalies into broader insights. For a cluster of 'STATUS_PENDING' claims in a single district, their AI-generated summary might state: "District X is showing a high concentration of 'STATUS_PENDING' claims, with an average delay of 45 days. This indicates a potential bottleneck in the payment disbursement process. Historical data analysis suggests this pattern has been consistent for the past six months, pointing towards a need for a policy review or procedural enhancement in this region." This explanation moves from the specific (a single claim) to the general (a district-wide trend), framing the issue in terms of efficiency and governance.

By implementing this tiered explanation system, the platform transcends its function as a mere data-matching tool. It becomes an intelligent assistant that empowers users at every level of the organization. The backend would store these pre-generated, role-specific explanations in a lookup table, keyed by `claim_id` and `role`. When the frontend

requests data for a specific user, it can then retrieve the appropriately customized explanation along with the reconciliation status, creating a seamless and highly personalized user experience. This strategy directly addresses the research goal of providing role-specific AI-generated explanations and demonstrates a deep understanding of user-centered design in the context of a complex financial system.

Frontend Dashboard Design: From Heat Maps to Actionable Insights

The frontend dashboard serves as the central nervous system of the reconciliation framework, translating raw data and algorithmic outputs into an intuitive, interactive, and actionable interface for users. Its design must prioritize clarity, immediacy, and workflow facilitation. The centerpiece of the dashboard, as specified, is the district-level heat map, which provides a high-level, geographical overview of the current reconciliation status across Nepal . This visualization must strictly reflect only the present state of reconciliation, avoiding the inclusion of predictive risk scores to maintain a focus on factual, verifiable conditions . The implementation of this map can be achieved using standard web mapping libraries such as Leaflet.js or D3.js, which are capable of rendering GeoJSON data for Nepal's districts and coloring them based on aggregated data [1](#) [26](#) . The color-coding scheme should be simple and universally understood: green for districts where claims are fully reconciled, yellow for districts with a mix of pending or mismatched claims, and red for districts with a significant number of unresolved discrepancies like missing payments. The data source for this map would be a server-side query that aggregates the reconciliation statuses of all claims associated with each district.

Complementing the heat map is a traffic-light styled data table that lists individual claims, allowing users to drill down from the macro view to the micro view. Each row in this table would correspond to a specific claim and be color-coded according to its reconciliation status: green for 'RECONCILED', yellow for 'AMOUNT_MISMATCH' or 'STATUS_PENDING', and red for 'MISSING_PAYMENT' . This table must be sortable and filterable, with district being a primary filter to allow users to isolate problems within a specific geographical area identified by the heat map. The most critical feature of this table is its interactivity. Clicking on a row should trigger a modal window or a dedicated panel that displays the detailed, role-specific AI-generated explanation for that particular anomaly, as discussed in the previous section. This creates a seamless transition from identifying a problem to understanding its nature. Alongside the explanation, the panel

should display key claim and payment details, such as the claimed amount versus the paid amount, the dates involved, and the names of the hospital and district.

Beyond passive visualization, the dashboard must empower users to take corrective action directly from the interface. This is where the concept of a "one-click resolution" becomes invaluable. For any flagged claim, a prominent button should be provided that generates a pre-filled dispute letter template . This feature leverages the rich contextual information already gathered by the system. The generated document would automatically populate fields with the claim ID, hospital name, the nature of the discrepancy ('AMOUNT_MISMATCH'), the exact figures involved, and the AI-generated explanation. This transforms a potentially arduous administrative task into a simple click, significantly reducing friction and encouraging timely resolution. The output can be rendered as a downloadable PDF or presented as a copyable text block, catering to different user preferences without over-engineering the solution . This workflow—from seeing a red dot on the map, drilling down to a red row in the table, reading the AI explanation, and clicking a button to initiate a resolution—forms the core value proposition of the user interface. It embodies the principle of turning data into decisions.

Finally, the entire interaction must be grounded in a robust audit trail. Every action taken within the system, from the initiation of a reconciliation run to the generation of a dispute letter, must be meticulously logged. This is not merely a technical requirement but a cornerstone of trust and governance [12](#) . The system should implement a simple append-only log table in the database, `audit_log`, with columns for `id`, `claim_id`, `action` (e.g., "Reconciliation Run," "Dispute Letter Generated"), `actor` (the user who performed the action), and `timestamp` . During a demonstration, showcasing the contents of this audit log serves as powerful proof of the system's commitment to transparency and accountability. It demonstrates that every step of the reconciliation and resolution process is documented, which is crucial for satisfying the expectations of auditors and compliance officers who oversee public funds [41](#) [57](#) . Even though this implementation may be basic (e.g., appending a row to a SQL table), the principle aligns with best practices for tamper-evident logging and data governance [18](#) [19](#) [45](#) . This combination of a clear geographical overview, an actionable data table, integrated AI explanations, and a transparent resolution workflow creates a comprehensive and powerful dashboard that directly addresses the user's research goal.

Governance and Accountability: The Role of Audit Trails and Predictive Heuristics

In the context of managing public funds within Nepal's social protection programs, governance and accountability are paramount. The reconciliation system, therefore, must be more than a technical tool for detecting mismatches; it must be architected as a pillar of financial integrity and fiduciary responsibility. Two critical components fulfill this role: a comprehensive audit trail that documents every action, and a predictive module that offers proactive risk assessment, albeit through a defensible and transparent methodology. The audit trail is the bedrock of accountability. In any system handling financial data, especially public funds, it is essential to know who did what and when ⁵⁵. The system must log all significant events in an immutable manner. This includes not only user-initiated actions like clicking a "resolve" button but also system-level events such as the start and completion of each reconciliation job. A simple yet effective implementation is an append-only log table in the database, `audit_log`, which records the `claim_id` involved, the nature of the `action` taken, the identity of the `actor` (user), and a precise `timestamp` for each event. This log provides an indisputable record of the reconciliation history for any given claim, which is invaluable for internal reviews, external audits by bodies like the World Bank's Audit Committee, and troubleshooting issues ^{41 58}. Demonstrating this log during a live demo serves as tangible evidence of the system's design for transparency and compliance, aligning with global principles of health data governance ¹².

While the core reconciliation engine is reactive—flagging problems that have already occurred—a predictive heuristic adds a crucial proactive dimension. This module does not purport to be a sophisticated machine learning model, as building and training one would be outside the project's scope and timeline. Instead, it should be positioned as a "rule-based risk scoring heuristic," a transparent and interpretable system for flagging potentially high-risk claims before they escalate ². This framing is strategically important; it manages stakeholder expectations by accurately representing the technology's capabilities, thereby avoiding the pitfalls of overstating AI sophistication that judges and mentors often penalize. The heuristic can be built upon a small set of common-sense rules derived from an understanding of financial fraud patterns and operational inefficiencies. For example, a risk score could be calculated based on factors such as whether a claim amount is an extreme outlier compared to a provider's historical average (e.g., more than three times the mean), whether the claim originates from a district known for payment delays, or whether the claim was submitted on a weekend, which might indicate rushed processing ³⁹. These rules provide a numerical risk score (e.g., from 0 to 100) that can be displayed alongside the reconciliation status in the

dashboard. A high-risk score would serve as an additional signal to an officer or ministry official, prompting them to investigate the claim with greater urgency. This approach combines the certainty of the reconciliation engine with the foresight of a simple analytical model, creating a more robust financial oversight tool. It moves the system from simply reporting errors to helping prevent them by directing investigative resources to the areas of highest perceived risk.

Strategic Positioning and Domain Alignment within Nepal's Health Financing Landscape

The ultimate success of this project depends not only on its technical execution but also on its ability to resonate with the strategic priorities of Nepal's social protection ecosystem. The system should be positioned not merely as a data reconciliation tool, but as a critical enabler of financial integrity and efficient resource allocation within the country's journey toward universal health coverage (UHC) [29](#). This reframing elevates its perceived value and aligns it with the objectives of key stakeholders, including the Social Security Fund (SSF), a pivotal government agency in this domain [31](#) [44](#), and international partners like the World Bank, which actively supports reforms in public financial management [8](#) [53](#). The core problem the system solves—discrepancies between approved claims and actual payments—is a direct challenge to the effective use of public funds, which are intended to protect households from financial catastrophe due to out-of-pocket medical expenses [52](#). By automating the detection of these discrepancies, the system acts as a safeguard, ensuring that contributions from formal sector workers and other funding streams flow efficiently from the SSF to healthcare providers, ultimately reaching the citizens who need them [16](#) [17](#).

The project's design choices are deeply informed by the institutional context of Nepal. The establishment of contributory social security schemes, managed by the SSF, represents a significant step in building a formal safety net for vulnerable populations [20](#) [21](#). The successful implementation of these schemes, such as the Employment Injury Scheme and the Sickness Benefit scheme, relies on a functional and transparent financial management information system (FMIS) [4](#) [13](#). The proposed reconciliation system directly strengthens the FMIS by providing a real-time monitoring and verification layer. This aligns with the World Bank's emphasis on Public Expenditure Review (PER) and Public Expenditure and Financial Accountability (PEFA) assessments, which are tools used to analyze health financing and public expenditure [53](#) [56](#). By creating a system that

tracks payments from source to destination, the project contributes to improving the transparency and accountability of health spending, a key theme in PEFA assessments. Furthermore, the system's focus on data governance principles, such as maintaining an immutable audit trail, mirrors the rigorous standards expected in projects funded by international financial institutions like the World Bank [12](#) [57](#). The system's architecture, which integrates with the open-source openIMIS platform, also supports the World Bank's recommendation to leverage existing government capacity and administrative structures, potentially reducing costs and fostering sustainability [5](#) [24](#). The ILO has also supported the SSF by customizing an openIMIS solution, highlighting the synergy between this project and ongoing efforts to digitize and strengthen social security administration in Nepal [40](#). By situating the reconciliation system within this broader narrative of institutional strengthening and fiscal responsibility, the project gains a strategic significance that transcends its technical function, positioning it as a valuable asset for enhancing the overall integrity and effectiveness of Nepal's social health protection program.

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